Hemoglobin_gene_comparision

it is bioinformatics project comparison of hemoglobin gene

sequence of human , mice, frog, and zebrafish.

species used:

homo sapiens (human)

mus musculus (mouse)

xenoplus laevis (frog)

danio rerio (zebrafish)

Data source

gene sequence were downloaded from NCBI databases in FASTA format.

Methods

1. downloded HBB sequence from NCBI.

2. store sequence in FASTA format .

3. perform multiple sequence alignment using CLUSTAL OMEGA.
4. calculate sequence similarity between species with help of global alignment (needleman wunsch algorithm)
5. construct phylogenetic tree by using MEGA.
6. made 3d protein structure using UCSF ChimeraX.
7. calculate G~C content%.
8. made summary pdf using CANVA.

Tools used

- [ ] FASTA dataset.
- [ ] Clustal Omega.
- [ ] MEGA.
- [ ] NCBI.
- [ ] UCSF ChimeraX
- [ ] Github.
- [ ] Canva.

Result

sequence similarity among species.

species 1	species 2	similarity %
human	mouse	82.10%
human	frog	68.40%
human	zebrafish	61.20%
mouse	frog	67.90%
mouse	zebrafish	60.50%
frog	zebrafish	65.80%

this tabel show evolutionary proximity through genetic similarity like human-mice have highest similarity due to mamalian ancestry.

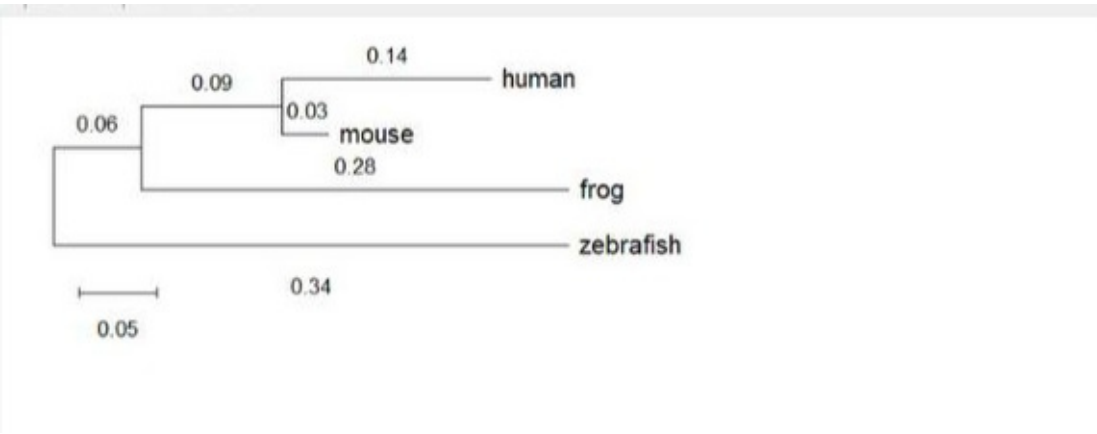
#while human-zebrafish low similarity due to early evolutionary divergence.

alignment sequence img.

- the HBB gene sequence from different species were aligned using Clustal Omega.
- the alignment shows how the sequences match across species.

- (*)means the amino acids are exactly the same in all sequences (highly conserved).
- (:) means the amino acids are different but have similar properties .
- (.) means the amino acid are slightly similar, while a blank space shows differences between sequences.

phylogentic tree.



biological interpretation

- **closest relative:** human and mice share most rcent common ancestry, shown by their close branching
- **Divergence:** zebrafish most diverged compare to mice from human due to early divergence of fish
- **evolutionary divergence:** the braching order reflects history of evolution zebrafish(fish) diverged first then frog(amphibian) lastly mammals (human and mice)
- **branch length:** represent genetic ditance like zebrafish show 0.34 units change which is highest mean most genetically distinct species

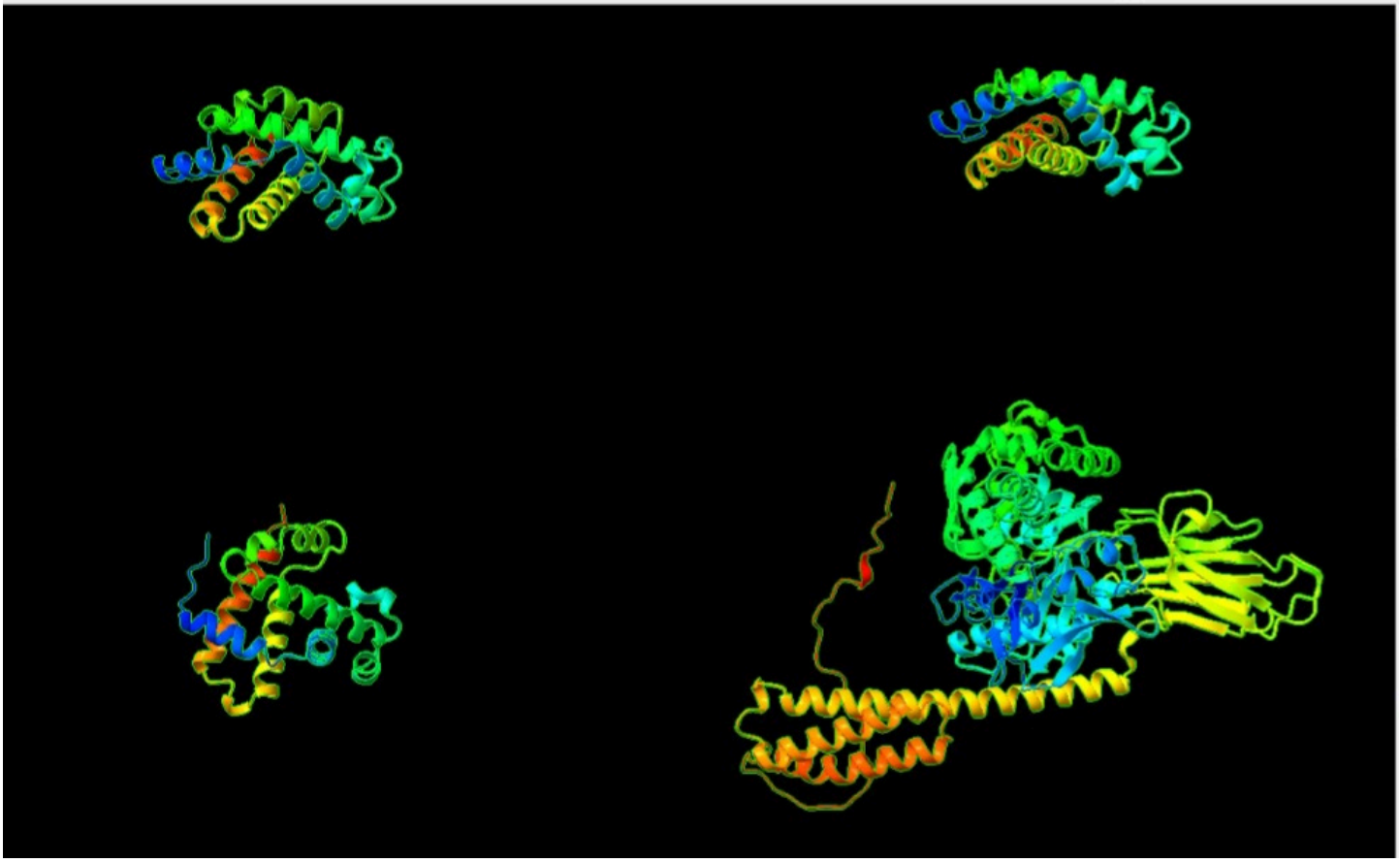
G~C content %

species	count of G	count of C	G~C %
human	325	320	40.1
mouse	161	160	51
frog	382	382	35.9
zebrafish	210	212	46.1

Here i measure Guanine and Cytosine percentage

- can be use this formula G+C/A+T+G+C ×100
- here i used [GC online calculator]

3d protein structure.



structure analysis

- The 3D structure of the HBB protein was created by **using UCSF ChimeraX**.
- The protein structures were downloaded from the AlphaFold Protein Structure Database using UniProt IDs.
- The structures were shown in ribbon form to see the shape of the protein.
- Images of the structures are compared for different species protein structure .

biological conclusion

- The results show that the HBB gene is highly conserved across different species.
- Human and mouse sequences are very similar, showing their close evolutionary relationship.
- Frog and zebrafish show more differences, which reflects greater evolutionary distance.
- This conservation indicates that the HBB gene is important for the function of hemoglobin and oxygen transport in vertebrates.
- Despite sequence variations, the overall protein structure and function remain largely preserved across vertebrates.